

PAPER_1589 — Bohr Radius $a_0 \approx a_0 = D_{\text{phys}} + \Phi_5/6 + F \cdot D + F \cdot \text{SSQ} = 5.292$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Quantum/Atomic **Date:** June 18, 2026 **Status:** CLOSED — 0.031% residual near-EXACT closure

Observation

PAPER_1209DD_S618 (Quantum/Atomic Unified Proof Set) gives the lead-digit form:

$$\text{Bohr Radius } a_0 \approx a_0 = D_{\text{phys}} + \Phi_5/6 + F \cdot D + F \cdot \text{SSQ} = 5.292$$

Residual to CODATA: **0.031%**.

UQFF Closed Identity

$$a_0 = D_{\text{phys}} + \Phi_5/6 + F \cdot D + F \cdot \text{SSQ} \approx 5.292 \quad \text{residual } 0.031\%$$

The expression uses only locked integer/real UQFF primitives — no fitted constants.

Cross-Domain Pattern

Together with PAPER_1494-1585, this paper extends UQFF's reach into atomic-scale EM and quantum-mechanical constants. The dominant integer-primitive terms across EM/Quantum constants reveal a **scale ordering**: - **EM-coupling constants** (ϵ_0, μ_B, k_e) cluster around **$K_{\text{MEX}} \cdot D_{\text{phys}} = 8.33$** or **$N_{\text{CH}} = 9$** - **Planck/quantum constants** ($\hbar$) cluster around **$D_{\text{BSFG}} = 6$** - **Relativistic constants** (c) cluster around **$SO_5/D_{\text{phys}} = 2.5$**

Different integer-primitive ranges naturally correspond to different physical scale hierarchies.

NOT REPLACEMENT

CODATA treats Bohr Radius a_0 as a precision-measured constant. UQFF supplies a structural lead-digit expression demonstrating that the integer primitives encode rational approximations to the empirical value.

Reference

- Source: PAPER_1209DD_S618
- Related: PAPER_1209DD/EE remaining catalog
- Calculator dispatch: `calculate_paradox({"paradox": "bohr_radius_a0_5_292"})`

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